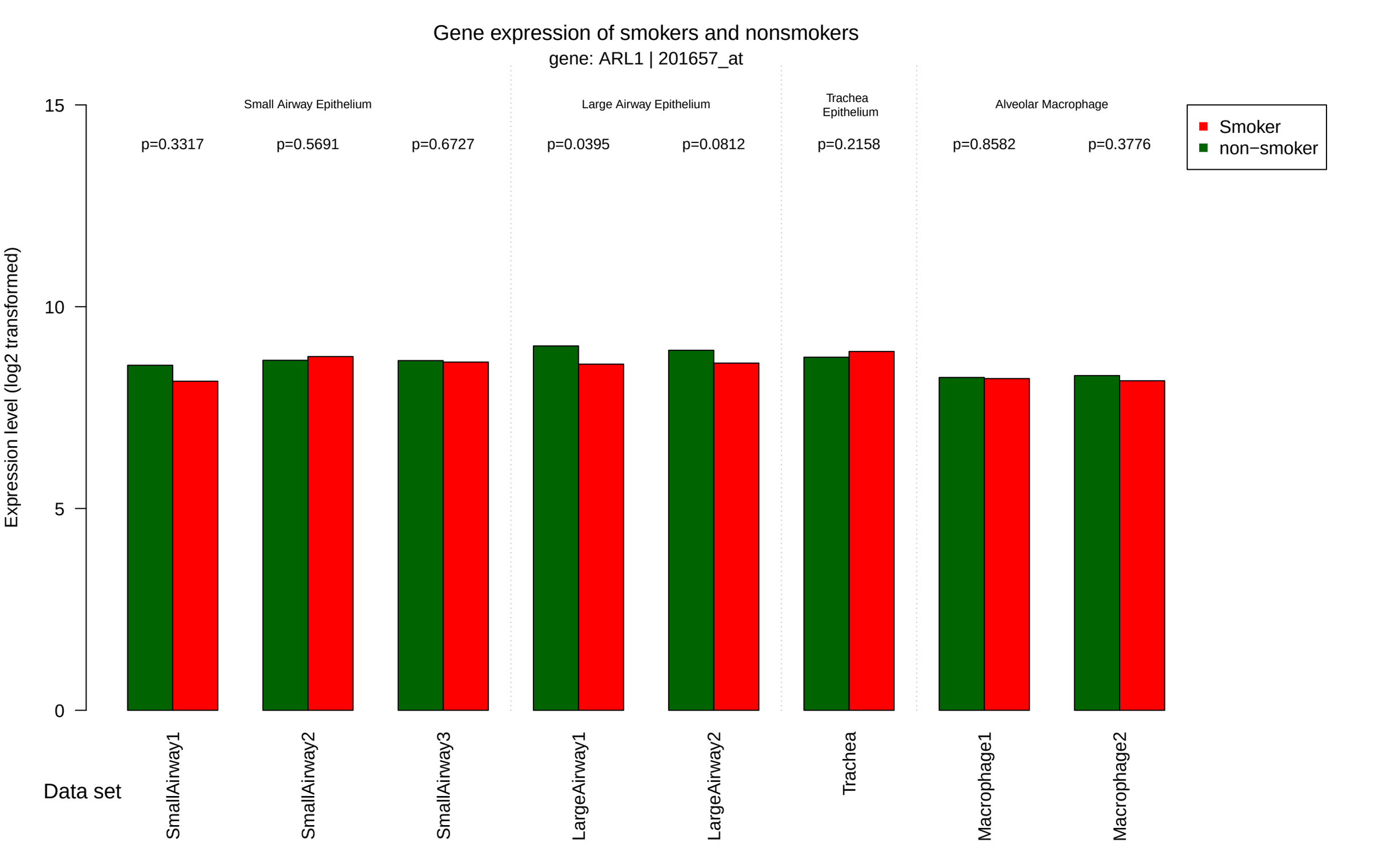
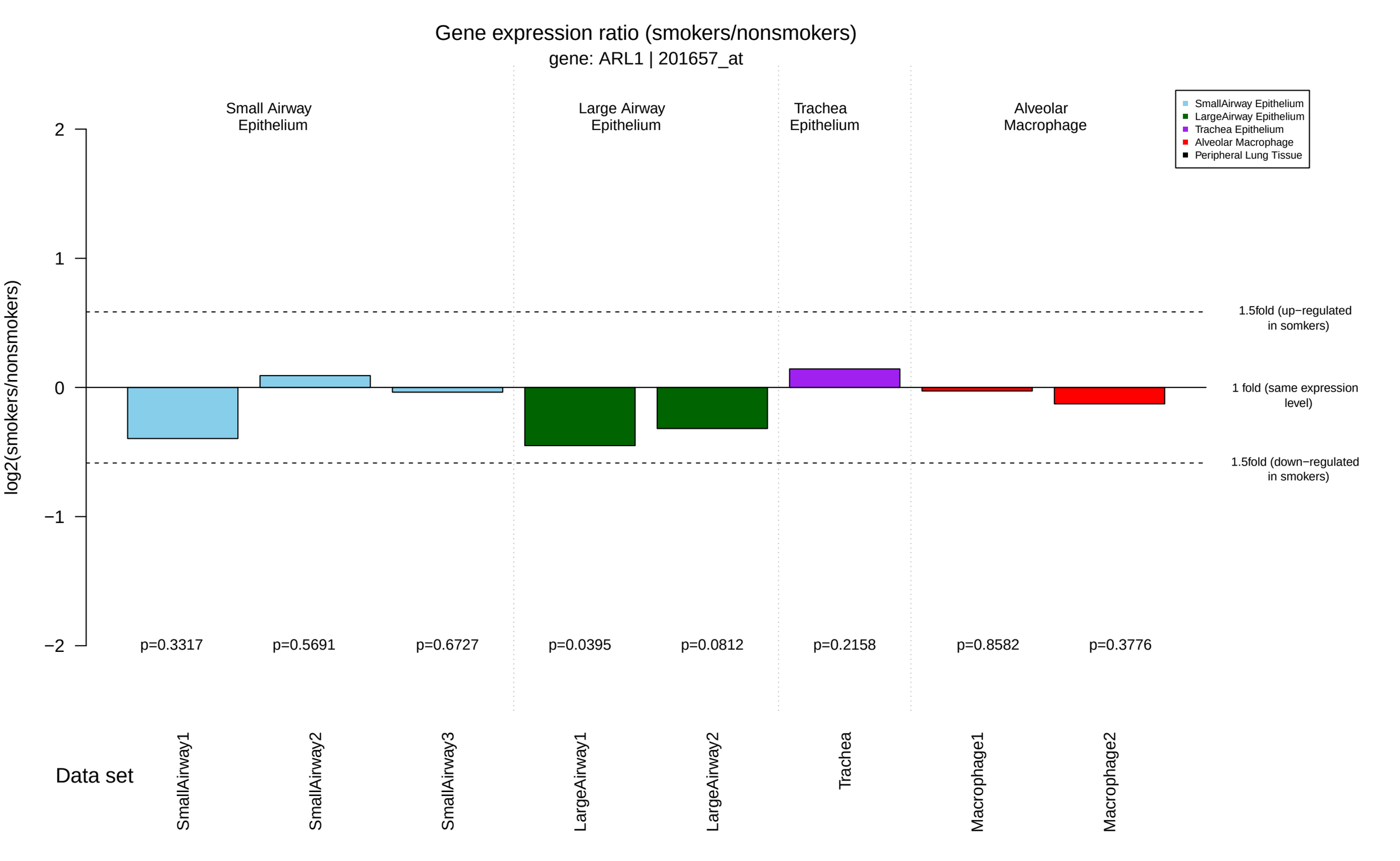
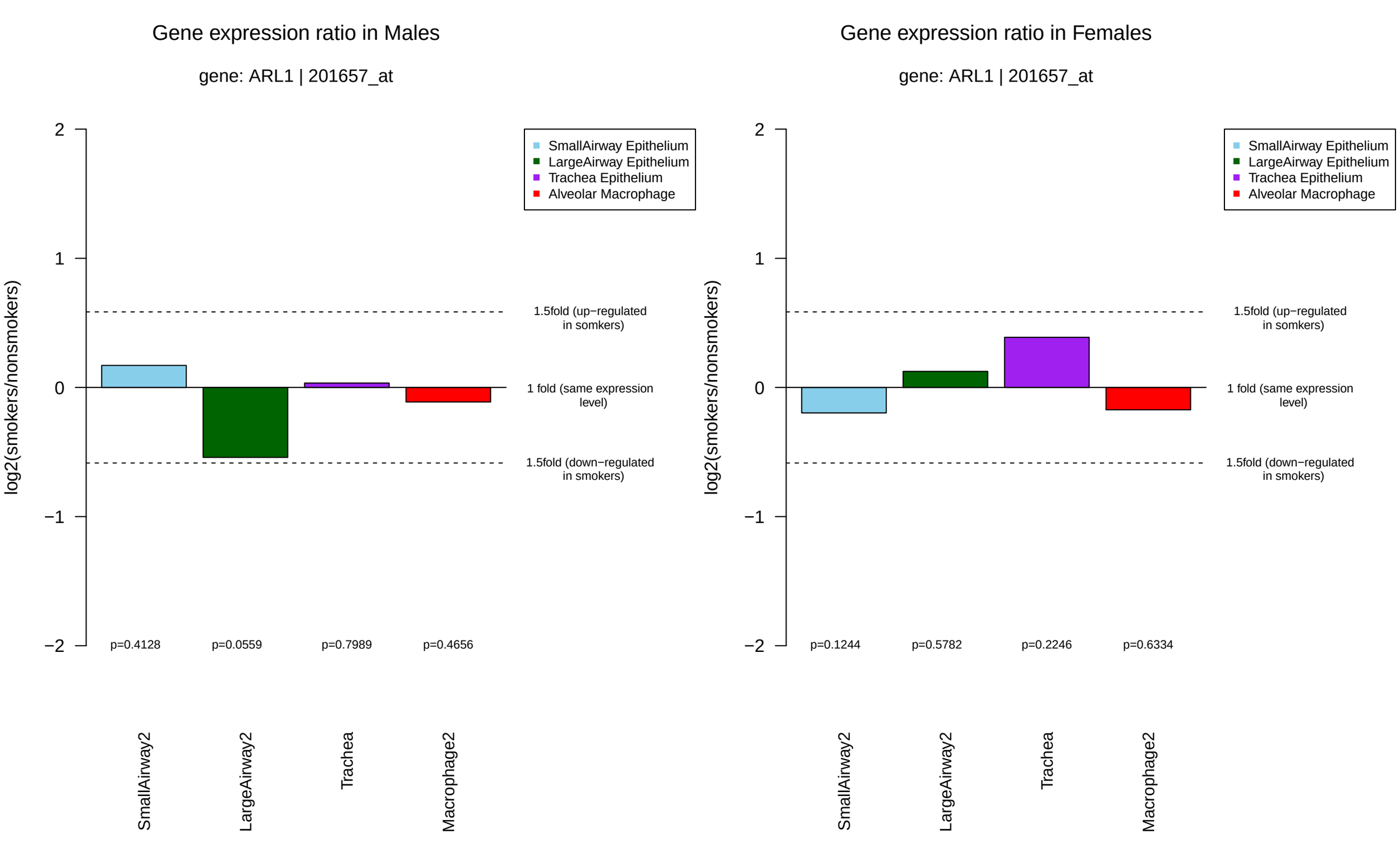
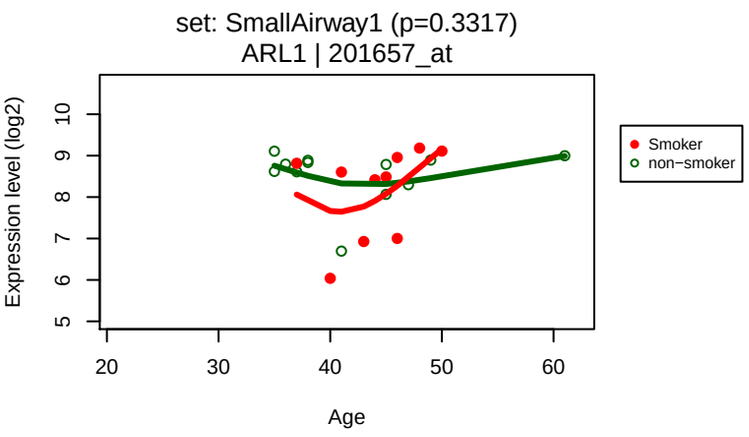
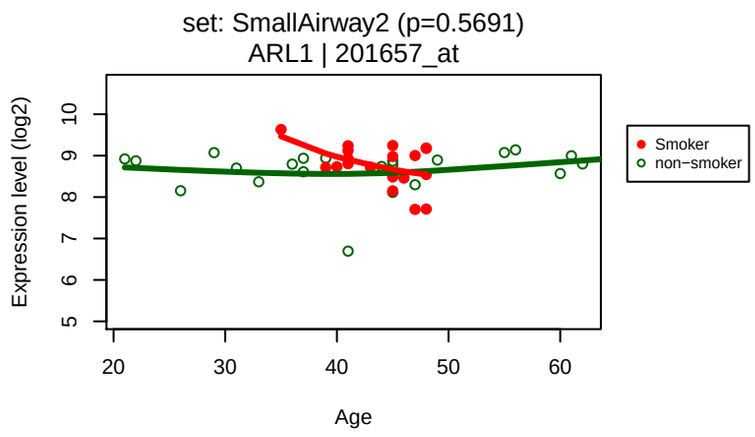
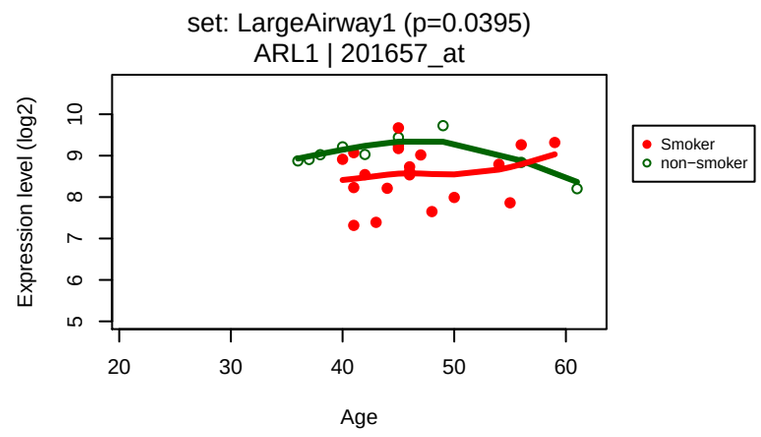

Gene expression of smokers and nonsmokers
gene: ARL1 | 201657_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.3317
p=0.5691
p=0.6727
p=0.0395
p=0.0812
p=0.2158
p=0.8582
p=0.3776
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

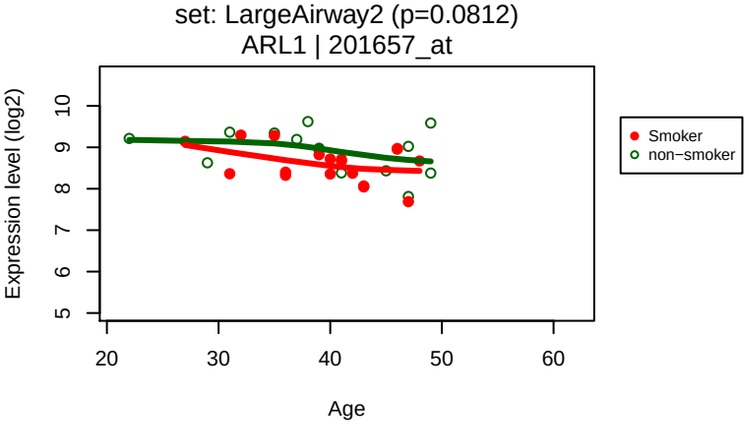

Gene expression ratio (smokers/nonsmokers)
gene: ARL1 | 201657_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3317
p=0.5691
p=0.6727
p=0.0395
p=0.0812
p=0.2158
p=0.8582
p=0.3776
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

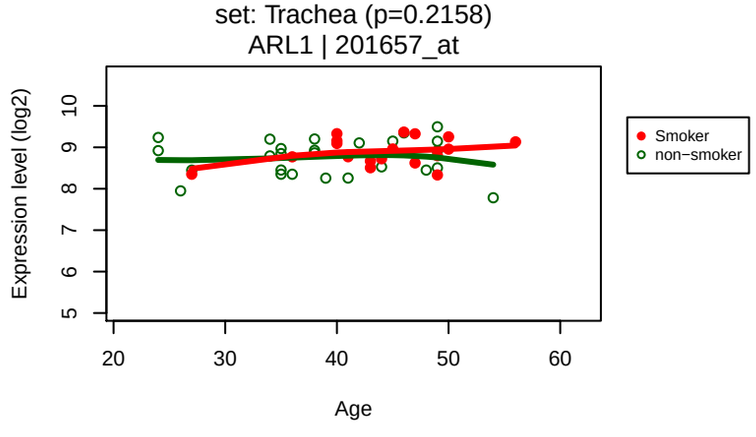

Gene expression ratio in Males
gene: ARL1 | 201657_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4128
p=0.0559
p=0.7989
p=0.4656
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ARL1 | 201657_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1244
p=0.5782
p=0.2246
p=0.6334
SmallAirway2
LargeAirway2
Trachea
Macrophage2

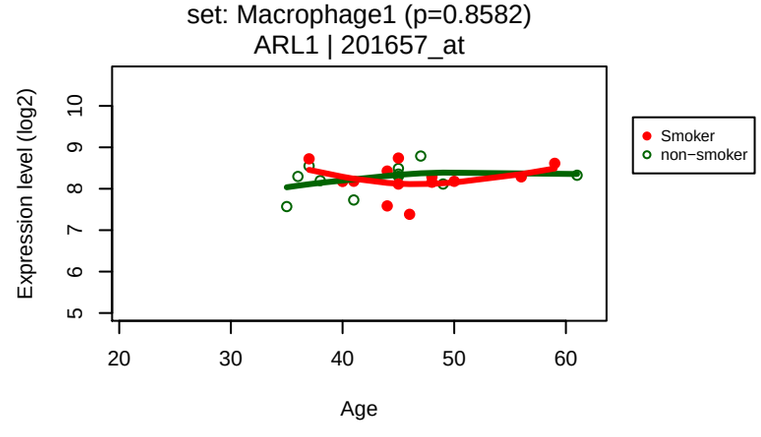

set: SmallAirway1 (p=0.3317)
ARL1 | 201657_at
Expression level (log2)
Age
Smoker
non−smoker

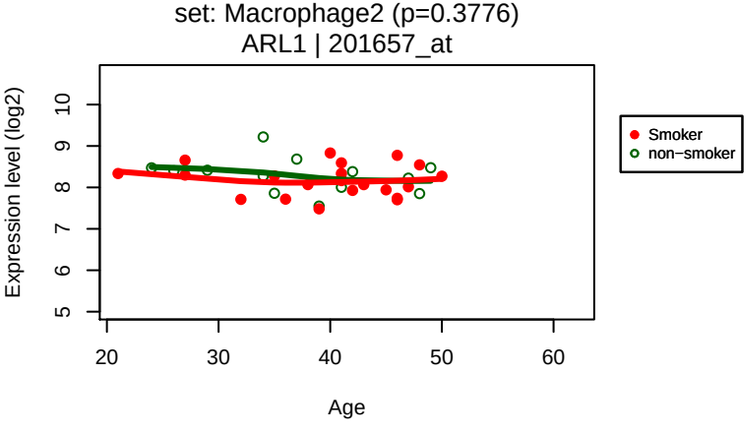

set: SmallAirway2 (p=0.5691)
ARL1 | 201657_at
Expression level (log2)
Age
Smoker
non−smoker

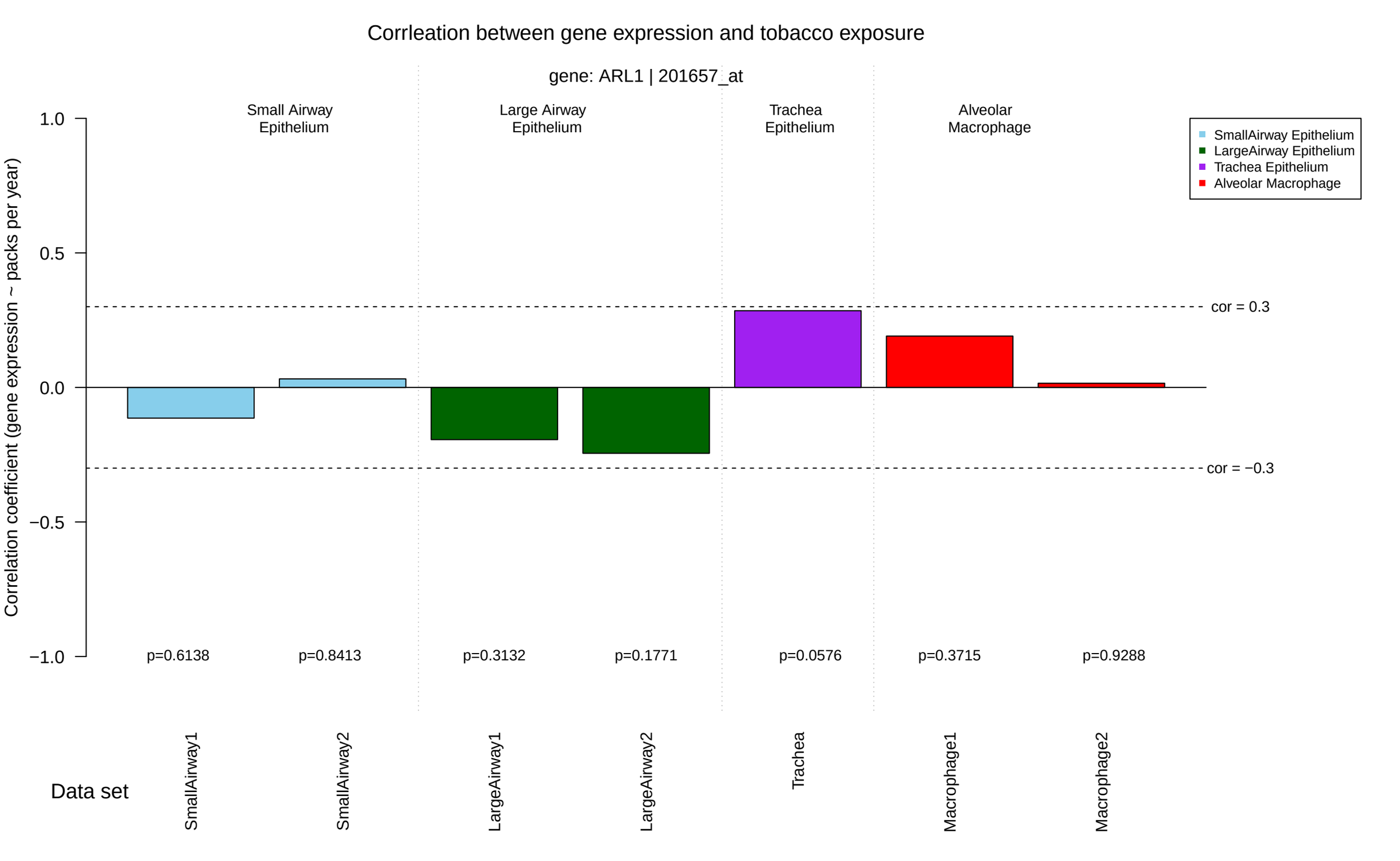

set: LargeAirway1 (p=0.0395)
ARL1 | 201657_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0812)
ARL1 | 201657_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.2158)
ARL1 | 201657_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.8582)
ARL1 | 201657_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.3776)
ARL1 | 201657_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ARL1 | 201657_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.6138
p=0.8413
p=0.3132
p=0.1771
p=0.0576
p=0.3715
p=0.9288
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2